09 - Docker, Prometheus and Grafana

Containers, metrics, dashboards, logs, alerts

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

docker-compose.yml; infrastructure/docker/*; dashboard/prometheus/*; dashboard/grafana/**; infrastructure/loki, promtail; src/metrics/exporter.py

Docker architecture

A single Compose project (fintech-research) with one bridge network (fintech-net) and six named volumes (pgdata, prometheus_data, loki_data, grafana_data, app_logs, promtail_positions). Host ports bind to 127.0.0.1 only. Every value has a default so 'docker compose up --build' works without a .env file.

Service	Role
postgres	results and metadata; schema applied from database/ at first start
worker	runs the pipeline, exposes /metrics on 8000, replays held-out data through the champion model
jupyter	JupyterLab; starts only after worker is healthy
prometheus	scrapes worker, prometheus, cadvisor, loki, grafana; loads alert rules
loki + promtail	log storage and shipping
cadvisor	container CPU, memory, network, disk
grafana	provisioned datasources, dashboards, alert rules

Verification state

Check	State
Compose file parsed	VERIFIED
Stack started with docker compose up	NOT VERIFIED
Dashboards viewed in a browser	NOT VERIFIED

Prometheus

Job	Targets	Purpose
worker	worker:8000	
prometheus	localhost:9090	
cadvisor	cadvisor:8080	
loki	loki:3100	
grafana	grafana:3000	

Global scrape interval: 5s. Alertmanager is intentionally not deployed; alerts are visible in the Prometheus and Grafana UIs.

Metrics exported by the Python worker (parsed from src/metrics/exporter.py)

Metric	Type	Help text	Labels
fintech_pipeline_ready	Gauge	1 once artefacts are built and metrics published	
fintech_pipeline_runs_total	Counter	Pipeline executions	status
fintech_pipeline_last_success_timestamp_seconds	Gauge	Unix time of last successful run	
fintech_pipeline_run_duration_seconds	Gauge	Duration of the last full run	
fintech_pipeline_stage_duration_seconds	Gauge	Duration of last execution of a stage	stage
fintech_pipeline_stage_success	Gauge	1 if the last execution of the stage succeeded	stage
fintech_pipeline_stage_rows	Gauge	Rows output by a stage	stage
fintech_pipeline_stage_errors_total	Counter	Stage failures	stage
fintech_db_write_failures_total	Counter	Failed PostgreSQL writes	table
fintech_db_rows_written	Gauge	Rows written to PostgreSQL in last run	table
fintech_data_validation_check_passed	Gauge	1 if the validation check passed	check
fintech_data_validation_failed_rows	Gauge	Rows violating a validation check	check
fintech_data_validation_pass_rate	Gauge	Share of validation checks passed	
fintech_data_cleaning_rows	Gauge	Row accounting during cleaning	kind
fintech_txn_dataset_transactions	Gauge	Transactions in processed dataset	channel
fintech_txn_dataset_frauds	Gauge	Labelled frauds in processed dataset	channel
fintech_txn_dataset_fraud_rate	Gauge	Overall labelled fraud rate	
fintech_txn_dataset_amount_total	Gauge	Total transaction value	kind
fintech_txn_fraud_rate_by_category	Gauge	Fraud rate per merchant category	merchant_category
fintech_txn_fraud_rate_by_hour	Gauge	Fraud rate per hour of day	hour
fintech_txn_frauds_by_pattern	Gauge	Frauds per generative pattern	pattern
fintech_txn_amount_quantile	Gauge	Amount quantiles	class, quantile
fintech_security_dataset_events	Gauge	Security events in dataset	event_type, severity
fintech_security_block_rate	Gauge	Share of attack events blocked	
fintech_security_attacks_by_source_country	Gauge	Attack events by source country	country
fintech_security_anomaly_roc_auc	Gauge	ROC-AUC of unsupervised detector vs known attacks	
fintech_security_anomalies_flagged	Gauge	Events flagged anomalous by Isolation Forest	
fintech_security_mean_response_seconds	Gauge	Mean response time	kind
fintech_model_metric	Gauge	Test-set evaluation metric	model, metric
fintech_model_confusion	Gauge	Confusion-matrix cell counts (test set)	model, cell
fintech_model_training_seconds	Gauge	Training + tuning wall time	model
fintech_model_threshold	Gauge	Decision threshold chosen on validation	model
fintech_model_cv_score	Gauge	Best cross-validated tuning score	model
fintech_model_champion	Gauge	1 for the champion model	model
fintech_risk_tier_transactions	Gauge	Test transactions per risk tier	tier
fintech_risk_tier_observed_fraud_rate	Gauge	Observed fraud rate per risk tier	tier
fintech_model_inference_seconds	Histogram	Batch inference latency	model

Metric	Type	Help text	Labels
fintech_trust_score_mean	Gauge	Mean Consumer Trust Score (0-100)	
fintech_trust_dimension_mean	Gauge	Mean score per trust dimension (0-100)	dimension
fintech_trust_segment_respondents	Gauge	Respondents per trust segment	segment
fintech_trust_platform_security_index	Gauge	Platform Security Index (0-100)	
fintech_trust_consumer_trust_index	Gauge	Combined Consumer Trust Index (0-100)	
fintech_trust_survey_respondents	Gauge	Survey respondents analysed	
fintech_trust_cronbach_alpha	Gauge	Cronbach's alpha	scale
fintech_trust_driver_correlation	Gauge	Spearman rho of trust score with driver	driver
fintech_trust_item_mean	Gauge	Mean Likert response per item (1-5)	item
fintech_trust_score_by_group	Gauge	Mean trust score by respondent group	group, value
fintech_trust_k_anonymity_min_group	Gauge	Smallest quasi-identifier group in published survey data	
fintech_stream_transactions_total	Counter	Replayed transactions scored	channel, predicted
fintech_stream_outcomes_total	Counter	Confusion outcome of replayed transactions	outcome
fintech_stream_amount_total	Counter	Value of replayed transactions	kind
fintech_stream_risk_tier_total	Counter	Replayed transactions per risk tier	tier
fintech_stream_security_events_total	Counter	Replayed security events	event_type, severity, blocked
fintech_stream_security_anomalies_total	Counter	Replayed security events flagged anomalous	
fintech_stream_last_batch_unixtime	Gauge	Unix time of last scored batch	

Two kinds: gauges describe the last pipeline run (values computed from data and models); counters and a histogram describe the live replay (outcomes, risk tiers, inference latency). Nothing is hardcoded: publish_summary() in the exporter reads the run summary produced by the pipeline.

Alert rules

Group	Rule	Expression
fintech-pipeline	WorkerDown	up{job="worker"} == 0
fintech-pipeline	PipelineStageFailed	fintech_pipeline_stage_success == 0
fintech-pipeline	DataValidationFailing	fintech_data_validation_pass_rate < 0.7
fintech-pipeline	DatabaseWriteFailures	increase(fintech_db_write_failures_total[15m]) > 0
fintech-models	fintech:stream_recall:ratio_rate10m	sum(increase(fintech_stream_outcomes_total{outcome="tp"}[10m])) / clamp_min(sum(increase(f
fintech-models	fintech:stream_precision:ratio_rate10m	sum(increase(fintech_stream_outcomes_total{outcome="tp"}[10m])) / clamp_min(sum(increase(f
fintech-models	LiveRecallDegraded	fintech:stream_recall:ratio_rate10m < 0.5
fintech-infrastructure	TargetDown	up == 0
fintech-infrastructure	ContainerHighMemory	container_memory_working_set_bytes{name=~"fintech-.+"} > 3e9

Grafana alert	Severity	For
Live fraud recall below 50%	warning	3m
Dataset fraud rate above 6%	critical	2m
Pipeline stage failed	critical	1m
Security anomaly surge	warning	2m

Grafana alert	Severity	For
Consumer Trust Index below 50	warning	5m

Loki and Promtail

config/logging.yaml writes JSON log lines (ts, level, logger, msg, optional stage/run_id/model) to logs/fintech.log. Promtail tails that file (shared app_logs volume), extracts level and logger as Loki labels, and pushes to Loki. Dashboards filter with stream selectors such as {job="fintech", level=~"WARNING|ERROR"}.

Grafana dashboards

Dashboard	Panels	Data sources	Refresh
Consumer Trust Indicators	12	grafana-postgresql-datasource, prometheus	30s
Cybersecurity Threat Monitoring	14	grafana-postgresql-datasource, loki, prometheus	10s
FinTech Research Overview	12	loki, prometheus	10s
Transaction & Fraud Analytics	18	grafana-postgresql-datasource, prometheus	10s
Machine Learning Model Performance	12	grafana-postgresql-datasource, prometheus	30s
Data Pipeline Monitoring	13	grafana-postgresql-datasource, loki, prometheus	10s
Docker Infrastructure Health	15	loki, prometheus	10s

Dashboards are generated by infrastructure/scripts/build_dashboards.py and provisioned from dashboard/grafana/dashboards (datasource UIDs prometheus, loki, postgres). Panels per dashboard:

Consumer Trust Indicators (consumer_trust.json)

Panel	Type	Source
Mean Consumer Trust Score	gauge	prometheus
Platform Security Index	gauge	prometheus
Consumer Trust Index (combined)	gauge	prometheus
Respondents / k-anonymity min group	stat	prometheus
Trust by dimension	bargauge	prometheus
Trust segments	piechart	prometheus
Trust score distribution (PostgreSQL)	histogram	grafana-postgresql-datasource
Mean Likert response per item (1-5)	bargauge	prometheus
Trust score by respondent group	bargauge	prometheus
Spearman correlation: trust vs driver	bargauge	prometheus
Cronbach's alpha (scale reliability)	bargauge	prometheus
Survey-derived trust per segment (PostgreSQL)	table	grafana-postgresql-datasource

Metrics queried: fintech_trust_consumer_trust_index, fintech_trust_cronbach_alpha, fintech_trust_dimension_mean, fintech_trust_driver_correlation, fintech_trust_item_mean, fintech_trust_k_anonymity_min_group, fintech_trust_platform_security_index, fintech_trust_score_by_group, fintech_trust_score_mean, fintech_trust_segment_respondents, fintech_trust_survey_respondents

Cybersecurity Threat Monitoring (cybersecurity_monitoring.json)

Panel	Type	Source
Attack events	stat	prometheus
Attack block rate	stat	prometheus
Anomaly detector ROC-AUC	stat	prometheus
Anomalies flagged	stat	prometheus
Mean response: attack (s)	stat	prometheus
Mean response: benign (s)	stat	prometheus
Attacks by type	bargauge	prometheus
Attack events by severity	piechart	prometheus
Attacks by source country	bargauge	prometheus
Daily attacks vs blocked	timeseries	grafana-postgresql-datasource
Security events / s by type	timeseries	prometheus
Anomalies flagged / s	timeseries	prometheus
Blocked share of replayed events	timeseries	prometheus
Validation failures and warnings	logs	loki

Metrics queried: fintech_security_anomalies_flagged, fintech_security_anomaly_roc_auc, fintech_security_attacks_by_source_country, fintech_security_block_rate, fintech_security_dataset_events, fintech_security_mean_response_seconds, fintech_stream_security_anomalies_total, fintech_stream_security_events_total

FinTech Research Overview (fintech_overview.json)

Panel	Type	Source
Pipeline ready	stat	prometheus
Last successful run	stat	prometheus
Fraud rate (dataset)	stat	prometheus
Champion PR-AUC	stat	prometheus
Mean trust score	stat	prometheus
Consumer Trust Index	stat	prometheus
Platform Security Index	stat	prometheus
Transactions scored (live)	stat	prometheus
Transactions scored / s	timeseries	prometheus
Security events / s by severity	timeseries	prometheus
Model PR-AUC (test set)	bargauge	prometheus
Warnings and errors	logs	loki

Metrics queried: fintech_model_champion, fintech_model_metric, fintech_pipeline_last_success_timestamp_seconds, fintech_pipeline_ready, fintech_stream_security_events_total, fintech_stream_transactions_total, fintech_trust_consumer_trust_index, fintech_trust_platform_security_index, fintech_trust_score_mean, fintech_txn_dataset_fraud_rate

Transaction & Fraud Analytics (fraud_detection.json)

Panel	Type	Source
Transactions	stat	prometheus
Labelled frauds	stat	prometheus

Panel	Type	Source
Fraud rate	stat	prometheus
Fraud value	stat	prometheus
Fraud share of value	stat	prometheus
Median amount fraud / legit	stat	prometheus
Fraud rate by merchant category	bargauge	prometheus
Fraud rate by hour of day	barchart	prometheus
Frauds by attack pattern	piechart	prometheus
Daily transactions vs frauds	timeseries	grafana-postgresql-datasource
Daily fraud value	timeseries	grafana-postgresql-datasource
Outcomes / s (TP, FP, FN, TN)	timeseries	prometheus
Live precision	stat	prometheus
Live recall	stat	prometheus
Fraud value caught vs missed / s	timeseries	prometheus
Live transactions by risk tier	piechart	prometheus
Observed fraud rate per tier (test set)	bargauge	prometheus
Risk tier summary (PostgreSQL)	table	grafana-postgresql-datasource

Metrics queried: fintech_risk_tier_observed_fraud_rate, fintech_stream_amount_total, fintech_stream_outcomes_total, fintech_stream_risk_tier_total, fintech_txn_amount_quantile, fintech_txn_dataset_amount_total, fintech_txn_dataset_fraud_rate, fintech_txn_dataset_frauds, fintech_txn_dataset_transactions, fintech_txn_fraud_rate_by_category, fintech_txn_fraud_rate_by_hour, fintech_txn_frauds_by_pattern

Machine Learning Model Performance (model_performance.json)

Panel	Type	Source
F1-score	bargauge	prometheus
Recall	bargauge	prometheus
Precision	bargauge	prometheus
ROC-AUC	bargauge	prometheus
PR-AUC (average precision)	bargauge	prometheus
False-positive rate (lower is better)	bargauge	prometheus
Confusion-matrix counts (test set)	table	prometheus
Training + tuning time (s)	bargauge	prometheus
Decision threshold (validation F1-optimal)	bargauge	prometheus
Inference latency p50 / p95 / p99 (per batch)	timeseries	prometheus
Live false-positive rate (replay)	timeseries	prometheus
model_results (latest run)	table	grafana-postgresql-datasource

Metrics queried: fintech_model_confusion, fintech_model_inference_seconds_bucket, fintech_model_metric, fintech_model_threshold, fintech_model_training_seconds, fintech_stream_outcomes_total

Data Pipeline Monitoring (pipeline_monitoring.json)

Panel	Type	Source
Runs (success / failure)	stat	prometheus
Last run duration	stat	prometheus
Validation pass rate	stat	prometheus
DB write failures	stat	prometheus
Stage duration (s)	bargauge	prometheus
Rows / items per stage	bargauge	prometheus
Stage success (1 = ok)	state-timeline	prometheus
Validation checks	table	prometheus
Cleaning row accounting	bargauge	prometheus
Rows written to PostgreSQL	bargauge	prometheus
Pipeline run history (PostgreSQL)	table	grafana-postgresql-datasource
Log volume by level	timeseries	loki
Pipeline stage logs	logs	loki

Metrics queried: fintech_data_cleaning_rows, fintech_data_validation_check_passed, fintech_data_validation_failed_rows, fintech_data_validation_pass_rate, fintech_db_rows_written, fintech_db_write_failures_total, fintech_pipeline_run_duration_seconds, fintech_pipeline_runs_total, fintech_pipeline_stage_duration_seconds, fintech_pipeline_stage_rows, fintech_pipeline_stage_success

Docker Infrastructure Health (system_performance.json)

Panel	Type	Source
Scrape targets up	stat	prometheus
Target status (1 = up)	state-timeline	prometheus
CPU usage (cores)	timeseries	prometheus
Memory working set	timeseries	prometheus
Network receive	timeseries	prometheus
Network transmit	timeseries	prometheus
Container uptime	bargauge	prometheus
Disk I/O (read + write)	timeseries	prometheus
Container restarts (last start time age)	bargauge	prometheus
Pipeline worker RSS memory	timeseries	prometheus
Prometheus ingested samples / s	timeseries	prometheus
Loki ingest rate (bytes/s)	timeseries	prometheus
Scrape duration by job	timeseries	prometheus
Grafana HTTP requests / s	timeseries	prometheus
Errors across stack (Loki)	logs	loki

No Prometheus metrics (SQL/log panels only).

Screenshots

No Grafana screenshots exist in the project (review/assets is empty). Capture them from the running stack and re-run this pack builder; none are fabricated.

PostgreSQL design

Table	Columns
pipeline_runs	run_id, started_at, finished_at, status, rows_raw, rows_clean, seed, data_hash, champion_model, notes
model_results	id, run_id, model, model_type, dataset, threshold, accuracy, precision, recall, f1, roc_auc, pr_auc, false_positive_rate, tn, created_at
model_metadata	id, run_id, model, artifact_path, params, n_features, threshold, train_seconds, cv_score
data_quality_checks	id, run_id, check_name, passed, failed_count, total, detail
transaction_daily_summary	id, run_id, day, n_transactions, n_frauds, amount_total, amount_fraud
security_daily_summary	id, run_id, day, n_events, n_attacks, n_blocked
risk_tier_summary	id, run_id, tier, transactions, frauds, amount, observed_fraud_rate, action
trust_respondent_scores	id, run_id, respondent_id, perceived_security, privacy_assurance, transparency, reliability, behavioural_intent, trust_score, segment
survey_responses	id, run_id, respondent_id, q01, q07, age_band, education, usage_frequency, uses_mfa, experienced_fraud, digital_literacy

Each table (except pipeline_runs) has run_id referencing pipeline_runs. Views v_latest_* (database/migrations/002_latest_run_views.sql) always point at the most recent successful run and feed the SQL panels. The pipeline falls back to SQLite when DATABASE_URL is not set.